
大开眼界

大开眼界1

计算：

$$2012 \times \frac{3\frac{3}{4} \times 1.3 + 3 \div 2\frac{2}{3}}{(1+3+5+7+9) \times 20 + 3}$$

$= 2012 \times \dfrac{\frac{15}{4}^{3} \times \frac{13}{10_{2}} + 3 \div \frac{8}{3}}{25 \times 20 + 3}$

$= 2012 \times \dfrac{\frac{39}{8} + 3 \times \frac{3}{8}}{503}$

$= 2012 \times \dfrac{\frac{39}{8} + \frac{9}{8}}{503}$

$= 2012 \times \dfrac{\frac{48}{8}^{6}}{503}$

$= 2012^{4} \times \dfrac{6}{503_{1}}$

$= 24$

大开眼界2

将下面的繁分数化简为最简分数。

$$1-\cfrac{9}{2+\cfrac{8}{3-\cfrac{7}{4-\cfrac{6}{5}}}}=\frac{1}{2}$$

$$=1-\cfrac{9}{2+\cfrac{8}{3-\cfrac{7}{\frac{14}{5}}}}$$

$$=1-\cfrac{9}{2+\cfrac{8}{3-\frac{5}{2}}}$$

$$=1-\cfrac{9}{2+\cfrac{8}{\frac{1}{2}}}$$

$$=1-\frac{9}{2+16}$$

$$=1-\frac{9}{18_2^{\,1}}$$

$$=\frac{1}{2}$$



本讲

大显身手

大显身手1

校内：【2023-2024济南期末】

计算下面各题．

(1) $12 \times \frac{8}{9} \div \frac{2}{3}$

$= 12 \times \frac{8}{9} \times \frac{3}{2}$

$= 16$

(2) $\frac{1}{3} + \frac{1}{3} \div \frac{4}{5}$

$= \frac{1}{3} + \frac{1}{3} \times \frac{5}{4}$

$= \frac{1}{3} + \frac{5}{12}$

$= \frac{4}{12} + \frac{5}{12}$

$= \frac{9}{12}$

$= \frac{3}{4}$

大显身手2

校内：【2023-2024北京期末】

计算下面各题，写出过程．

(1) $\frac{5}{6}\times\frac{3}{7}\div\frac{2}{5}+\frac{2}{7}$

$=\frac{5}{\cancel{6}_{2}}\times\frac{\cancel{3}^{1}}{7}\div\frac{2}{5}+\frac{2}{7}$

$=\frac{5}{14}\div\frac{2}{5}+\frac{2}{7}$

$=\frac{5}{14}\times\frac{5}{2}+\frac{2}{7}$

$=\frac{25}{28}+\frac{2}{7}$

$=\frac{25}{28}+\frac{8}{28}$

$=\frac{33}{28}$

(2) $\frac{3}{8}\div\frac{1}{3}-\frac{1}{8}$

$=\frac{3}{8}\times\frac{3}{1}-\frac{1}{8}$

$=\frac{9}{8}-\frac{1}{8}$

$=\frac{8}{8}$

$=1$

大显身手3

把下列小数转化为分数，分数转化为小数。

(1)	0.7	0.875	(3)	0.6	(4) 0.75	7.3	2.5
	$\frac{7}{10}$	(2) $\frac{7}{8}$		$\frac{3}{5}$	$\frac{3}{4}$	(5) $\frac{73}{10}$ 或 $7\frac{3}{10}$	(6) $\frac{5}{2}$ 或 $2\frac{1}{2}$

(1) $7 \div 10 = 0.7$

(2) $0.875 = 875 \div 1000 = \frac{875}{1000} = \frac{7}{8}$

(3) $3 \div 5 = 0.6$

(4) $3 \div 4 = 0.75$

(5) $7.3 = 73 \div 10 = \frac{73}{10}$

(6) $2.5 = 25 \div 10 = \frac{25}{10} = \frac{5}{2}$

大显身手 4

计算：

(1) $\frac{4}{9} \times 6.5 - \frac{7}{18}$

$= \frac{4}{9} \times \frac{13}{2} - \frac{7}{18}$

$= \frac{26}{9} - \frac{7}{18}$

$= \frac{52}{18} - \frac{7}{18}$

$= \frac{45}{18}$

$= \frac{5}{2}$

(2) $2.5 \times \frac{2}{5} + 9.63 - 2.1 \div \frac{1}{3}$

$= 2.5 \times 0.4 + 9.63 - 2.1 \times 3$

$= 1 + 9.63 - 6.3$

$= 10.63 - 6.3$

$= 4.33$


本讲
融会
贯通

融会贯通 1

计算：

$$\frac{7}{18}\times\frac{6}{7}+\frac{5}{18}$$

$$=\frac{\cancel{7}^{1}}{\cancel{18}_{3}}\times\frac{\cancel{6}^{1}}{\cancel{7}_{1}}+\frac{5}{18}$$

$$=\frac{1}{3}+\frac{5}{18}$$

$$=\frac{6}{18}+\frac{5}{18}$$

$$=\frac{11}{18}$$

$$\frac{4}{11}\times\frac{11}{13}-\frac{2}{9}\div\frac{13}{3}$$

$$=\frac{\cancel{4}^{1}}{\cancel{11}_{1}}\times\frac{\cancel{11}^{1}}{13}-\frac{2}{9}\div\frac{13}{3}$$

$$=\frac{4}{13}-\frac{2}{9}\div\frac{13}{3}$$

$$=\frac{4}{13}-\frac{2}{\cancel{9}_{3}}\times\frac{\cancel{3}^{1}}{13}$$

$$=\frac{4}{13}-\frac{2}{39}$$

$$=\frac{12}{39}-\frac{2}{39}$$

$$=\frac{10}{39}$$

融会贯通2

校内：【2023-2024河南期末】

计算：

$$2-\frac{6}{13}\div\frac{9}{26}+\frac{2}{3}$$

$=2-\frac{6}{13}\times\frac{26}{9}+\frac{2}{3}$

$=2-\frac{4}{3}+\frac{2}{3}$

$=\frac{6}{3}-\frac{4}{3}+\frac{2}{3}$

$=\frac{4}{3}$ 或 $1\frac{1}{3}$

$$\left(\frac{7}{6}-\frac{5}{9}\right)\div\frac{11}{18}\div 11$$

$=\left(\frac{21}{18}-\frac{10}{18}\right)\times\frac{18}{11}\times\frac{1}{11}$

$=\frac{1}{\cancel{18}}\times\frac{\cancel{18}}{11}\times\frac{1}{11}$

$=\frac{1}{11}$

融会贯通3

计算：

$\frac{17}{18}+\frac{8}{9}\times 0.4$

$=\frac{17}{18}+\frac{8}{9}\times\frac{2}{5}$

$=\frac{17}{18}+\frac{16}{45}$

$=\frac{85}{90}+\frac{32}{90}$

$=\frac{117}{90}=\frac{13}{10}$

$=1\frac{3}{10}$ 或 1.3

$\frac{9}{14}\div 0.4-\frac{6}{7}$

$=\frac{9}{14}\div\frac{2}{5}-\frac{6}{7}$

$=\frac{9}{14}\times\frac{5}{2}-\frac{6}{7}$

$=\frac{45}{28}-\frac{6}{7}$

$=\frac{45}{28}-\frac{24}{28}$

$=\frac{21}{28}=\frac{3}{4}$

$\frac{9}{25}\div\frac{9}{19}\times 0.5$

$=\frac{9}{25}\div\frac{9}{19}\times\frac{1}{2}$

$=\frac{9}{25}\times\frac{19}{9}\times\frac{1}{2}$

$=\frac{19}{50}$

融会贯通 4

校内：【2023-2024湖南期末】

计算：

$$1\frac{2}{7} \div 1\frac{13}{14} + \frac{1}{3}$$

$= \frac{9}{7} \div \frac{27}{14} + \frac{1}{3}$

$= \frac{9^{1}}{7_{1}} \times \frac{14^{2}}{27_{3}} + \frac{1}{3}$

$= \frac{2}{3} + \frac{1}{3}$

$= \frac{3}{3}$

$= 1$

融会贯通5

校内：【2025湖南XSC】

计算：

(1) $3\frac{2}{5} \div 2\frac{1}{8} + 5\frac{2}{5} - 0.125$

$= \frac{17}{5} \div \frac{17}{8} + \frac{27}{5} - \frac{1}{8}$

$= \frac{17}{5} \times \frac{8}{17} + \frac{27}{5} - \frac{1}{8}$

$= \frac{8}{5} + \frac{27}{5} - \frac{1}{8}$

$= \frac{35}{5} - \frac{1}{8}$

$= 7 - \frac{1}{8}$

$= 6\frac{7}{8}$

(2) $\frac{1}{24} \div \left(\frac{1}{3} - \frac{1}{4} + \frac{1}{12}\right)$

$= \frac{1}{24} \div \left(\frac{4}{12} - \frac{3}{12} + \frac{1}{12}\right)$

$= \frac{1}{24} \div \frac{2}{12}$

$= \frac{1}{24} \times \frac{6}{2}$

$= \frac{1}{4}$

融会贯通6

校内：【2023重庆XSC】

计算：

$$1.65 \div \left(\frac{1}{4}+0.8\right)-\left(\frac{1}{2}+\frac{1}{3}\right) \times \frac{24}{35} .$$

$= \frac{33}{20} \div \left(\frac{1}{4}+\frac{4}{5}\right)-\left(\frac{1}{2}+\frac{1}{3}\right) \times \frac{24}{35}$

$= \frac{33}{20} \div \left(\frac{5}{20}+\frac{16}{20}\right)-\left(\frac{3}{6}+\frac{2}{6}\right) \times \frac{24}{35}$

$= \frac{33}{20} \div \frac{21}{20}-\frac{5}{6} \times \frac{24}{35}$

$= \frac{33}{20} \times \frac{20}{21}-\frac{5}{6} \times \frac{24}{35}$

$= \frac{11}{7}-\frac{4}{7}$

$= \frac{7}{7}$

$= 1$

融会贯通7

校内：【2024陕西西安XSC】

$$2-\left[1\frac{1}{2}\times\left(2\frac{1}{2}\div3\frac{1}{3}\right)\right]\times\frac{2}{3}$$

$=2-\left[\frac{3}{2}\times\left(\frac{5}{2}\div\frac{10}{3}\right)\right]\times\frac{2}{3}$

$=2-\left[\frac{3}{2}\times\left(\frac{5^{1}}{2}\times\frac{3}{\cancel{10}_{2}}\right)\right]\times\frac{2}{3}$

$=2-\left[\frac{3}{2}\times\frac{3}{4}\right]\times\frac{2}{3}$

$=2-\frac{\cancel{9}^{3}}{\cancel{8}_{4}}\times\frac{\cancel{2}^{1}}{\cancel{3}_{1}}$

$=2-\frac{3}{4}$

$=1\frac{1}{4}$

融会贯通8

校内：【2023陕西西安XSC】

$$\frac{3}{4}\times\left[\left(1\frac{1}{2}+2\frac{2}{3}\right)\div\frac{5}{6}-2\frac{1}{3}\right]$$

$=\frac{3}{4}\times\left[\left(\frac{3}{2}+\frac{8}{3}\right)\div\frac{5}{6}-\frac{7}{3}\right]$

$=\frac{3}{4}\times\left[\left(\frac{9}{6}+\frac{16}{6}\right)\div\frac{5}{6}-\frac{7}{3}\right]$

$=\frac{3}{4}\times\left[\frac{25}{6}\div\frac{5}{6}-\frac{7}{3}\right]$

$=\frac{3}{4}\times\left[\frac{25}{6}\times\frac{6}{5}-\frac{7}{3}\right)$

$=\frac{3}{4}\times\left[5-\frac{7}{3}\right]$

$=\frac{3}{4}\times\frac{8}{3}$

$=2$